

Unequal Job Opportunities and Well-Being Inequality

A Latent-Jobs Structural Decomposition

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Observed

C_i

Gini of consumption

single adults 0.2633 couples 0.2268

Well-being

M_i

Gini of the attained-bundle money metric

single adults 0.2498 couples 0.1974

low earnings $\Leftarrow$ tastes **or** opportunities **or** resources

equivalised, household-weighted, within each population

Compensation

unequal job opportunities should not become
unequal well-being

Responsibility

preference-based choices should not be
compensated

Compensation vs. Responsibility

$\nexists W : \text{Full Compensation} \wedge \text{Full Responsibility}$

[Fleurbaey & Maniquet]

Research question

How much inequality in money-metric well-being is associated with unequal job opportunities rather than heterogeneous preferences, once labour supply is modelled as choice among latent jobs?

1. Do observed labour-supply choices reflect preferences alone, or also heterogeneous latent job opportunities?
2. How do conclusions differ when welfare evaluates the attained bundle versus the ex-ante opportunity prospect?
3. In a restricted P/A/B decomposition, how much welfare inequality is associated with preferences, local access, and earning opportunities?

Latent jobs

Aaberge, Dagsvik & Strøm

Dagsvik & Jia

Capéau, Decoster & Dekkers

Money metrics

Fleurbaey & Maniquet

Decoster & Haan

Jacquet, Jia & Thoresen

Decomposition

Shorrocks

Creedy & Hérault

Mühlhan

reforms and *changes* → a *level* of well-being inequality

preferences vs. opportunities, inside an estimated latent-jobs model

$j = (e, k, h, w)$

4 occupation groups

$C_i(j)$ priced through EUROMOD

$$v_i(j) = L_i(j) + \beta_c \log\left(\frac{C_i(j)}{\lambda_c}\right)$$

L_i

value of leisure: age; children for
women

β_c

weight on log consumption,
estimated

λ_c

units convention

Opportunities: the choice probability

$$P_i(j) = \frac{\exp\{v_i(j)\} g_i(j)}{\int \exp\{v_i(k)\} g_i(k) d\nu(k)}$$

access

employment mass

hours

bands

occupation

by sex

wage offers

education, experience

valued or plentiful

EU-SILC France

collected 2016, incomes 2015

1,540 single-adult households

2,223 couple households

EUROMOD

French 2015 policy system

$h \in [5, 70]$ $w \in [2, 590]$

100 priced alternatives per household

Household-specific opportunities

free coordinates 41

population fit 0.0140

Common opportunities

free coordinates 10 / 16

population fit 0.0273 / 0.0264

criterion worse by at least 141.64

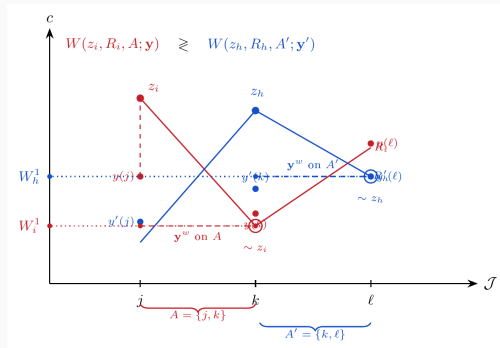
choices reflect opportunities, not preferences alone

single adults; population fit is the mean absolute deviation of predicted from observed moments

Selected estimates

	Single adults	Couples
β_c consumption weight	2.0387 (0.2917)	2.1017 (0.2939)
$\beta_{E,s}$ local unemployment exposure, in access	-1.4422 (0.2356)	-1.1924 (0.1557)
β_{wH} high education, wage-offer location	0.1491 (0.0308)	0.1817 (0.0197)
σ log-wage dispersion	0.3815 (0.0133)	0.3631 (0.0072)

cluster-robust standard errors in parentheses; the two populations are estimated separately



Own-set equal-consumption equivalents: each individual gets a common consumption on every job in their own ability set; the level at which the preferred job is indifferent to the attained bundle is their money metric, comparable across individuals. Adapted from Haydar and Maniquet (2026), work in progress.

$$J_i = \int e^{L_i(j)} \left(\frac{C_i(j)}{\lambda_c} \right)^{\beta_c} \widehat{g}_i(j) d\nu(j), \quad H_i = \int e^{L_i(j)} \widehat{g}_i(j) d\nu(j),$$

$$M_i^{\text{EA}} = \lambda_c \exp \left\{ \frac{\log J_i - \log H_i}{\beta_c} \right\}$$

J_i

the actual prospect

H_i

same jobs, equal pay

M_i^{EA}

flat-consumption equivalent

$$M_i^{\text{att}} = C_i^{\text{obs}} \exp \left\{ \frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c} \right\}$$

o

non-employment, reachable by all

j_i^{obs}

the only route for opportunities

Inequality and Shapley decomposition

P

preferences: age, children

A

local unemployment
exposure, region, urban
or rural location, year

B

wage-offer location

held fixed

household resources,
needs and composition

$$\phi_k^P = \sum_{S \subseteq \{P, A, B\} \setminus \{k\}} \frac{|S|! (3 - |S| - 1)!}{3!} \left[v^P(S \cup \{k\}) - v^P(S) \right]$$

$$v^P(S) = I^P(\emptyset) - I^P(S) \quad I^P: \text{household-weighted Gini of } M^P$$

How much is associated with opportunities?

Ex-ante prospect

equivalised

Single adults

20.3%

Couples

7.9%

Attained bundle

equivalised

raw 14.8%

1.9%

raw 21.3%

3.5%

raw 2.4%

raw 6.7%

A + B as % of the relevant baseline Gini

Household resources, needs and composition held fixed · A is the currently estimated local-access channel

Which opportunity channel matters?

Single adults

access
15.5%

earning opportunities

4.8%

$A + B = 20.3\%$

Couples

access
3.7%

earning opportunities

4.2%

$A + B = 7.9\%$

ex-ante prospect, equivalised: % of baseline Gini

single adults: access > earnings

couples: earnings > access, close

attained bundle: earnings > access for both populations

Where does baseline inequality go?

shares of baseline Gini · ex-ante prospect, equalised · current restricted P/A/B game · A = current local-access channel

		Single adults	Couples
Preferences	ϕ_P/l_0	0.1%	-1.8%
Local access	ϕ_A/l_0	15.5%	3.7%
Earning opportunities	ϕ_B/l_0	4.8%	4.2%
Not allocated by this exercise	$1 - (\phi_P + \phi_A + \phi_B)/l_0$	79.6%	93.9%
Total		100.0%	100.0%

Couples: A + B alone = 7.9%; P + A + B = 6.1% because the preference contribution is negative (-1.8%).

Not allocated is not a channel: it holds household resources, needs and composition fixed, retains sex-specific preference and opportunity blocks, and excludes wage-draw luck and the common offer spread, which exceed the location differences equalised.

one decimal; the not-allocated row closes each column to the total

What is not claimed

Scope

Household resources, needs and composition
held fixed

Access: local access, not total opportunity

Preliminary decomposition

Not claimed

Not causal

No parameter uncertainty yet

Preferences are not responsibility

Neither perspective is designated primary

prospects versus attained outcomes: two different welfare questions

1. Choices reflect opportunities, not preferences alone.
2. Prospects: access leads for single adults. Attained outcomes: earnings lead. Couples: no reversal.
3. Restricted P/A/B: a bounded part of well-being inequality; preference sign not robust.

next: resources and needs as a fourth channel · parameter uncertainty · validation

Appendix map

Model	Utility specification	Opportunity density	Likelihood	Identification
Estimation	Preferences	Employment access	Hours and occupation	Wage offers
	Common opportunities	Inference		
Fit	Extensive margin	Fit by margin	Thirty-seven-hour point	Matched households
	Marginal rate of substitution			
Welfare	ATT derivation	EA derivation	EA validation	Equivalisation
Decomposition	Central figure	EA Gini points	ATT Gini points	EA channel shares
	ATT channel shares	Composition within P/A/B	Shapley vs one-factor	Not allocated
	Robustness			
Data	Sample selection	EUROMOD pricing	Numerical design	

Backup: utility specification

$$v_i(j) = L_i(j) + \beta_c \log\left(\frac{C_i(j)}{\lambda_c}\right), \quad L_i(j) = \beta_\ell^g(\mathbf{x}_i) \mathcal{B}(\tilde{\ell}_i(j); \theta_\ell^g)$$

$$\tilde{\ell}_i(j) = \frac{80 - h}{10}, \quad \mathcal{B}(z; \theta) = \frac{z^\theta - 1}{\theta}$$

$\beta_\ell^g(\mathbf{x}_i)$

intercept, centred age and its
square (units of 10 years)

k_i

children under 20, women only

couples

sum of the two spouses' leisure
terms

Backup: opportunity density

$$g_i(j) = g_i^E \cdot \left(g_i^H(h) g_i^{\text{Occ}}(k) g_i^W(w | k) \right)^{E_i(j)}$$

access

g_i^E : local unemployment exposure, region, urban or rural location, year

hours

g_i^H : elevated bands over a residual set

occupation

g_i^{Occ} : sex-specific group masses

wage offers

g_i^W : log-normal, truncated to [2, 590]

wage-offer location $\mu_i(k)$: education, potential experience and its square, occupation

Backup: sampled-alternative likelihood

$$V_{ij} = v_i(j) + \log g_i(j) - \log q_{ij}$$

$$\Pr(y \mid \mathcal{C}_i) = \frac{n_y \exp V_{iy}}{\sum_{s \in \mathcal{C}_i} \exp V_{is}}$$

q_{ij}

proposal density, fitted out of fold

$\mathcal{C}_i$

observed job and 100 draws: 101
rows

n_y

multiplicity of a repeated draw

Backup: identification

Functional form

smooth preferences in hours;
banded hours opportunities

Exclusion

local unemployment exposure and
geography shift access, not utility

Independence

wage offers independent of hours,
given occupation

maintained, not tested · not a causal design

Backup: preference parameters

	Single men	Single women	Couple men	Couple women
leisure weight, intercept	8.5199 (3.5480)	5.8683 (2.1346)	3.9814 (0.7466)	13.1029 (4.7760)
leisure weight, age	1.4820 (1.0236)	0.0678 (0.4792)	-0.0080 (0.0287)	-0.1228 (0.1260)
leisure weight, age squared	0.7783 (0.7732)	1.0000 (at bound)	0.0065 (0.0030)	0.0073 (0.0103)
leisure weight, children	—	0.1666 (0.4422)	—	-0.3055 (1.1472)
leisure curvature θ_ℓ	-1.6263 (0.3301)	-0.9274 (0.2161)	-0.9761 (0.1357)	-1.6863 (0.2458)
consumption weight β_c	2.0387 (0.2917)		2.1017 (0.2939)	

cluster-robust standard errors; the male children term in couples is structurally zero

Backup: employment-access parameters

	Singles	Couples
employment constant	−3.1735 (0.4053)	men −2.1233 (0.3195); women −2.8148 (0.3021)
local unemployment exposure	−1.4422 (0.2356)	−1.1924 (0.1557)
densely populated	−0.0288 (0.2209)	−0.1823 (0.1657)
intermediate density	0.0641 (0.2614)	−0.4174 (0.1861)
Region 2	−0.3780 (0.3292)	−0.1546 (0.2442)
Region 3	−0.1117 (0.3891)	0.0873 (0.2808)
Region 4	−0.8218 (0.3810)	0.0150 (0.3068)
Region 5	−0.5162 (0.3274)	−0.1455 (0.2555)
Region 6	−0.7222 (0.3528)	−0.2844 (0.2748)
Region 7	−0.5374 (0.3503)	−0.1396 (0.2651)
Region 8	−0.4512 (0.3401)	−0.1659 (0.2660)

references: the first region and thinly populated areas; couples share the shifters across spouses

Backup: hours and occupation opportunities

Hours bands				
	Singles	Couple men	Couple women	
part-time lower	0.0298	−1.1094	−0.5173	
part-time upper	0.7080	0.0835	1.1586	
narrow full-time	2.0656	2.2888	2.0284	
full-time upper	1.9387	2.3801	1.6825	
long hours	−0.0834	0.6977	−0.3109	
Occupation masses				
	Singles		Couples	
	men	women	men	women
Occupation 2	−1.3737	0.1135	−1.5097	0.2074
Occupation 3	−2.1096	−0.4060	−2.2532	−0.1804
Occupation 4	−0.3259	0.5632	0.1594	0.8159

estimates only; standard errors in the paper. References: the residual hours set and the craft and elementary occupation group

Backup: wage-offer equation

	Singles	Couples
intercept	2.0138 (0.0561)	2.0480 (0.0335)
low education	0.0566 (0.0372)	−0.0433 (0.0217)
high education	0.1491 (0.0308)	0.1817 (0.0197)
potential experience	0.2408 (0.0862)	0.5643 (0.0546)
experience squared	−0.0282 (0.0390)	−0.1598 (0.0243)
Occupation 2	−0.0347 (0.0399)	−0.0740 (0.0231)
Occupation 3	0.0622 (0.0383)	0.0394 (0.0228)
Occupation 4	0.2785 (0.0368)	0.2146 (0.0223)
log-wage dispersion σ	0.3815 (0.0133)	0.3631 (0.0072)

medium education is the reference; experience in units of 20 years

Backup: common versus household-specific opportunities

Single adults	free coordinates	criterion	population fit
household-specific opportunities	41	6253.463	0.0140
common opportunity distribution	10	6403.974	0.0273
common shape; employment and hours in utility	16	6395.108	0.0264

same households, sampled alternatives and criterion; not converted into a formal test

Single adults

free coordinates 41, interior 40

criterion 6253.463

smallest Hessian eigenvalue 0.1669

Couples

free coordinates 47, interior 47

criterion 10283.034

smallest Hessian eigenvalue 0.0771

cluster-robust standard errors on the household · eigenvalues on the interior block

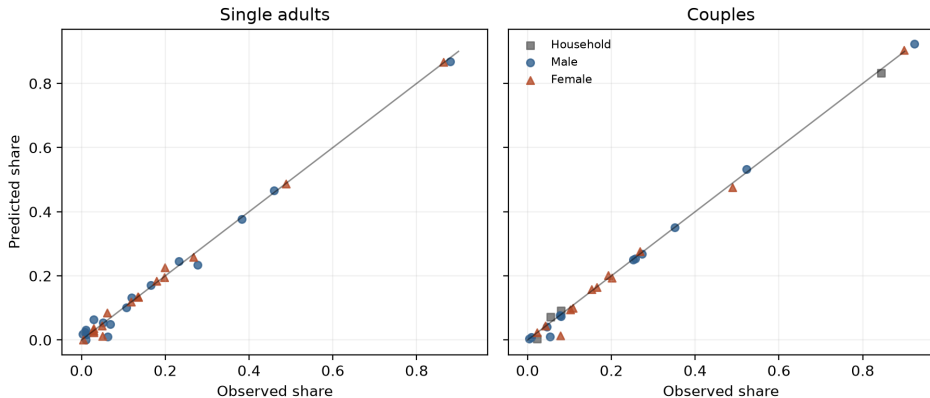
Backup: extensive-margin fit

	weighted accuracy	model-simulated range	integration ratio
single women	85.7%	80.0–85.9%	0.096
coupled women	89.8%	88.1–91.2%	0.049
single men	withheld	—	0.361
coupled men	withheld	—	0.353

withheld where numerical integration error is too large relative to sampling variation

Backup: fit by margin

Corrected population moments: observed versus predicted



Backup: the thirty-seven-hour point

single men

5.2

pp

single women

3.8

pp

coupled men

4.4

pp

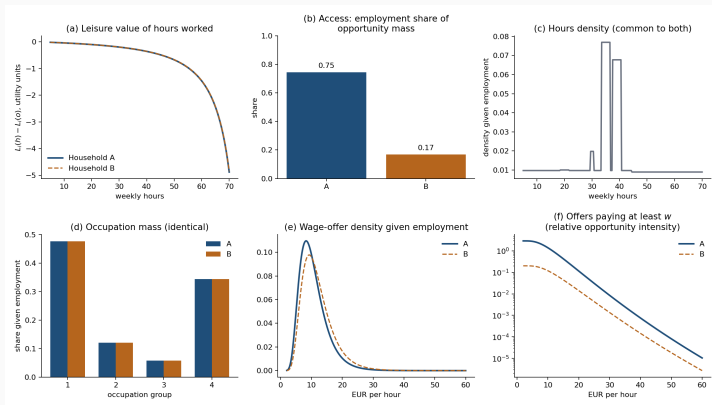
coupled women

6.5

pp

observed minus predicted share at the observed thirty-seven-hour mass point

Backup: matched households

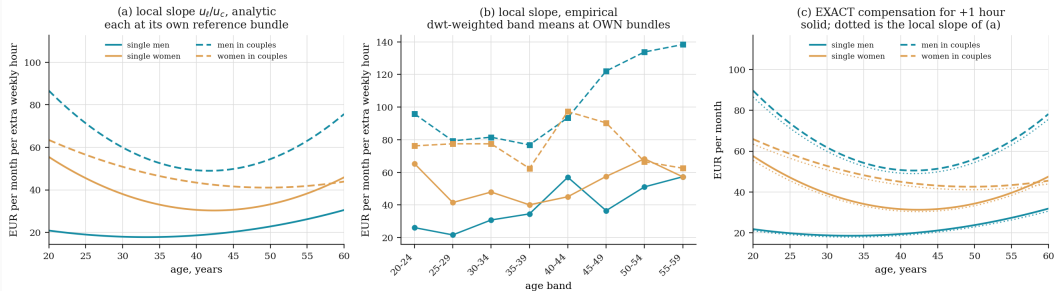


access: $A = 14.5 \times B$

wage-offer location: $B + 11.5$ log points

Backup: marginal rate of substitution

The local compensation u_l/u_c by age and sex, and the exact compensation for one more weekly hour



Backup: attained-bundle derivation

$$L_i(o) + \beta_c \log\left(\frac{M_i^{\text{att}}}{\lambda_c}\right) = L_i(j_i^{\text{obs}}) + \beta_c \log\left(\frac{C_i^{\text{obs}}}{\lambda_c}\right)$$

$$\implies M_i^{\text{att}} = C_i^{\text{obs}} \exp\left\{\frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c}\right\}$$

Reference job

the preferred job when every job pays the same:
here non-employment for every household

Staying-home equivalent

a verified property of this specification, not a
general theorem

Backup: ex-ante derivation

$$J_i = \int_{\Omega_i} e^{L_i(j)} \left(\frac{C_i(j)}{\lambda_c} \right)^{\beta_c} \widehat{g}_i(j) d\nu(j)$$

$$J_i^{\text{ref}}(m) = \left(\frac{m}{\lambda_c} \right)^{\beta_c} H_i, \quad H_i = \int_{\Omega_i} e^{L_i(j)} \widehat{g}_i(j) d\nu(j)$$

$$J_i^{\text{ref}}(M_i^{\text{EA}}) = J_i \quad \implies \quad M_i^{\text{EA}} = \lambda_c \exp \left\{ \frac{\log J_i - \log H_i}{\beta_c} \right\}$$

a common rescaling of $\widehat{g}_i$ cancels $\cdot \log J_i$: expected utility of the best reachable job, up to a constant

Backup: ex-ante numerical validation

Identities

constant-consumption identity

one-state identity

common intensity-rescaling invariance

flat-reference inversion

exact Shapley adding-up

Numerics

exact reference integral on the priced set

effective sample size and weight concentration

two-seed stability of welfare, Gini and Shapley

finite positive welfare for every household

independent reimplementations

all passed for both populations · a correct computation, not causal identification or parameter uncertainty

Backup: equivalisation

$$M_i^{p,eq} = \frac{M_i^p}{e_i}, \quad e_i : \text{modified-OECD scale}$$

Couples, ex ante: A + B

21.3% $\longrightarrow$ 7.9%

raw $\rightarrow$ equivalised; % of baseline Gini

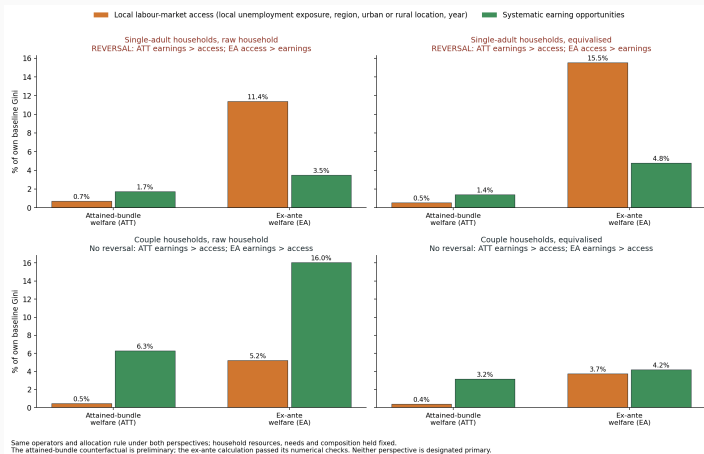
Couples, ex ante: preferences

0.00291 $\longrightarrow$ -0.00212

Gini points, raw $\rightarrow$ equivalised

a normative convention, reported both ways; populations never combined or compared

Backup: central figure



Backup: ex-ante decomposition in Gini points

	scale	baseline Gini	P	A	B	A + B, % of baseline	ordering
Single adults	raw	0.1595	0.00157	0.01812	0.00555	14.8	access > earnings
	equivalised	0.1406	0.00020	0.02183	0.00673	20.3	access > earnings
Couples	raw	0.1277	0.00291	0.00667	0.02049	21.3	earnings > access
	equivalised	0.1200	-0.00212	0.00449	0.00504	7.9	earnings > access

exact Shapley contributions in Gini points; household-weighted

Backup: attained-bundle decomposition in Gini points

	scale	baseline Gini	P	A	B	A + B, % of baseline	ordering
Single adults	raw	0.2337	0.0012	0.0016	0.0040	2.4	earnings > access
	equivalised	0.2451	-0.0003	0.0013	0.0034	1.9	earnings > access
Couples	raw	0.2035	0.0064	0.0009	0.0128	6.7	earnings > access
	equivalised	0.1971	-0.0019	0.0007	0.0062	3.5	earnings > access

exact Shapley contributions in Gini points; preliminary; household-weighted

Backup: ex-ante channels, % of baseline

		P				A		B	
	scale	I_0	Gini pts	% of I_0	Gini pts	% of I_0	Gini pts	% of I_0	
Single adults	raw	0.1595	0.00157	0.99	0.01812	11.37	0.00555	3.48	
	equivalised	0.1406	0.00020	0.14	0.02183	15.52	0.00673	4.79	
Couples	raw	0.1277	0.00291	2.28	0.00667	5.22	0.02049	16.04	
	equivalised	0.1200	−0.00212	−1.76	0.00449	3.75	0.00504	4.20	

each contribution divided by its own baseline Gini I_0 ; signed; the rows are not forced to sum to one hundred, because the remainder is not allocated by this exercise

Backup: attained-bundle channels, % of baseline

		P			A		B	
	scale	I_0	Gini pts	% of I_0	Gini pts	% of I_0	Gini pts	% of I_0
Single adults	raw	0.2337	0.0012	0.53	0.0016	0.69	0.0040	1.73
	equivalised	0.2451	-0.0003	-0.13	0.0013	0.54	0.0034	1.40
Couples	raw	0.2035	0.0064	3.16	0.0009	0.45	0.0128	6.28
	equivalised	0.1971	-0.0019	-0.98	0.0007	0.37	0.0062	3.15

each contribution divided by its own baseline Gini I_0 ; signed; preliminary

Backup: composition of the explained component

$$s_k = \frac{\phi_k}{\phi_P + \phi_A + \phi_B}, \quad k \in \{P, A, B\}$$

ex-ante prospect, equivalised	Single adults	Couples
Preferences s_P	0.7%	−28.5%
Local access s_A	75.9%	60.6%
Earning opportunities s_B	23.4%	67.9%
Total	100.0%	100.0%

These are shares of the currently explained P/A/B change, not shares of total inequality. Negative components are possible under Shapley attribution.

Backup: Shapley versus one-factor equalisation

$$\text{one-factor: } v^P(\{k\}) = I^P(\emptyset) - I^P(\{k\})$$

$$\text{Shapley: } \phi_k^P = \text{average over all orders of } v^P(S \cup \{k\}) - v^P(S)$$

$$\phi_P^P + \phi_A^P + \phi_B^P = I^P(\emptyset) - I^P(\{P, A, B\})$$

one-factor effects need not add up · Shapley contributions do, exactly

Backup: what is not allocated

Held fixed

household resources, needs and
composition

Retained

sex- and household-type-specific
preference and opportunity blocks

Not equalised

wage-draw luck and the common
offer spread

wage-offer location differences: at most about 0.10 log points

common offer spread: about 0.37 log points

next: household resources, needs and composition as a fourth channel D; $A + B$ stays the opportunity component

Backup: robustness of the attained-bundle decomposition

Second simulation seed

every coalition Gini and contribution
reproduces closely, including the preference
sign change

Observed job in the simulation

attained under every coalition by 2.9% of single
adults and 3.9% of couples
excluding it moves the total change by at most
9.3%, with no sign change

simulation checks, not confidence intervals

Backup: sample selection

screen	single-adult	couple
the raw France 2016 input file	4,038	5,965
one- or two-adult households (composition screen)	4,038	5,965
every adult aged 20 to 60	2,131	3,662
no adult still in education	2,036	3,521
no old-age, disability or survivor pension receipt	1,755	3,218
labour status in scope	1,598	2,412
no other employable or earning member	1,564	2,323
hours and wage inside the modelled support	1,555	2,275
hours outside [5,70] or unsupported military occupation (ISCO 0) on an employed decider	1,543	2,223
observed chosen alternative priced at non-positive disposable consumption	1,540	2,223
estimation sample	1,540	2,223

Backup: EUROMOD pricing

Inputs

a work arrangement: hours, hourly wage, earnings split; household roster and non-labour inputs

EUROMOD

French 2015 policy system; only deciders receive counterfactual overrides

Output

$C_i(j)$: household disposable income, summed over all members

every sampled alternative and every welfare integration point is priced
tax-benefit accounting identity holds to machine precision

Estimation

101

rows per household: the observed
job and 100 sampled alternatives

Predictive fit

2,048

common integration points per
household

Supports

[5, 70]

weekly hours; wages in [2, 590] per
hour

sampling devices are computational: they are not offers and enter no welfare measure